

Tyler Kuipers

SOFTWARE ARCHITECT

Edmonton, Canada

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About Me

I've been writing software for about 10 years, and the last three I've spent as a Software Architect doing mostly backend and API work. I own four REST APIs that other teams and external partners build against, so a lot of my time goes to contracts, versioning, and the standards that keep them from breaking as they grow. I came up on the JVM: Java at Lockheed Martin, then several years of Gosu, which is close enough to Java that moving between them is mostly muscle memory. OO design and the usual design patterns are still how I reach for structure. I also like the less glamorous work, taking a brittle legacy component and rebuilding it into something reliable. I'm comfortable in AWS and Kubernetes and lean on Postgres, and I put real weight on tests and code review, since that's what lets a small team ship quickly without a mess of regressions.

Work Experience

Alberta Motor Association

SOFTWARE ARCHITECT

Edmonton, Alberta

Jan 2023 - Present

- Own 4 REST APIs serving 5 internal teams and external partners. Designed the contracts and versioning, approve new endpoint additions, and maintain OpenAPI design standards in Confluence. (Python, REST, OpenAPI, TypeScript)
- Led a 2-3 engineer rewrite of a legacy document generation system into containerized microservices on EKS. 4-5 services (with sidecars) on a 3-5 node cluster, deployed with Helm. Wired up Datadog metrics for failure rate, throughput, generation time, and storage time, replacing a system that had frequent outages. (Kubernetes, EKS, Docker, Helm, Datadog, C#)
- Set technical standards for a 12-person team through code reviews, RFC authorship, and PR checklists / API design guides. Member of the Architecture Steering Committee. Help newer developers navigate legacy codebases and trained the team on TDD methodology.
- Architected the infrastructure for Canada's first pay-as-you-go insurance product across multiple AWS accounts. 10 Lambda functions, 99.9% uptime since launch, 2.8M insured trips and 76,637 invoices processed. Won the 2024 Guidewire Innovation Award (1 of 3 worldwide). (Python, AWS Lambda, Terraform, TypeScript)
- Built a self-service developer platform solo that took environment provisioning from 2-3 weeks down to 2 hours (60x). Self-service create/destroy, database snapshots, and pull-from-prod, adopted by 15 people across 3 teams and used to spin up 120+ environments. It made production debugging routine: a cancellation bug nobody could reproduce got root-caused in days using snapshotted production data. (Python, Terraform, AWS, Jenkins, Docker)
- Built an event-driven pipeline that ingests an external partner's policy data: S3 ingestion, a Kubernetes CronJob, a translation service, and transactional calls to a composite API, with OAuth 2.0 auth and Datadog monitoring. (Python, Kubernetes, AWS S3, OAuth 2.0)

Arterys Inc. (Acquired by Tempus Labs)

SOFTWARE DEVELOPER

Calgary, Alberta (Remote)

Aug 2021 - Aug 2022

- Built and maintained backend services that ran medical AI algorithms and returned results to the platform. (TypeScript, Node.js, AWS)
- Improved AI algorithm processing turnaround time using Redis caching and AWS infrastructure optimizations.
- Monitored production systems via Elasticsearch and Kibana, catching and resolving issues before customer impact.

Alberta Motor Association

SOFTWARE DEVELOPER | SOFTWARE TEAM LEAD

Edmonton, Alberta

Jun 2018 - Feb 2021 | Feb 2021 - Aug 2021

- Designed and built automated test framework that reduced defects by 30% on subsequent projects. Adopted across the insurance engineering team (15 developers / all Guidewire projects) and still in use 4+ years later. (Gosu/Java)
- Led 5-6 person development team on Guidewire insurance platform using TDD methodology. Performed Scrum Master role. Delivered projects on time and budget with no critical defects post-launch. Code reviews and mentoring junior developers.
- Led Git migration for 12-15 developers transitioning from SVN. Created Docker image for local Guidewire development environments. Trained 5-6 developers on TDD methodologies through hosted training sessions.

Pleasant Solutions

SOFTWARE DEVELOPER

Edmonton, Alberta

Jan 2018 - Jun 2018

- Built HMI for a concrete plant control system using TDD methodology. Daily work in C# / .NET, heavily utilizing the MVC design pattern, with REST integration and tests written with NUnit and Moq. (C#, .NET, MVC, REST)

Lockheed Martin Canada

ASSOCIATE SOFTWARE ENGINEER

Halifax, Nova Scotia

Jan 2017 - Jan 2018

- Contributed to Data Fusion module for Combat Management Systems, combining data from multiple underwater and above-water sensor sources. (Java, SQL)
- Designed and implemented scenario recorder system that enabled re-running failed test scenarios at will, improving debugging and testing capabilities.

- Built web app for spatial/temporal data visualization (Edmonton River Atlas). (PHP, JavaScript, PostgreSQL, PostGIS)
- Wrote undergrad thesis on spatial-temporal referencing methods.
- Presented research through poster presentations.

Education

The Open University MSC. COMPUTING (SOFTWARE ENGINEERING) Currently pursuing Master's in Computing at The Open University.	Milton Keynes, England (Remote) May 2024 - Ongoing
The King's University B.S. IN COMPUTER SCIENCE	Edmonton, Alberta Sep 2012 - Dec 2016

Skills

Backend & API Design	I own four REST APIs that internal teams and external partners build against, so most of my attention goes to the contract: clean versioning, sane error handling, and design standards that hold up as the surface area grows. I came up on the JVM and still reach for OO design and the standard patterns when I'm structuring a service. Designing backends that other people depend on, and not surprising them, is the part of the job I care most about.
Quality & Release Engineering	Earlier in my career I designed an automated test framework that cut defects by 30% on later projects; the insurance engineering team (15 developers) is still using it four years on. I'm a long-time TDD practitioner and I do a lot of code review, since catching a problem there is far cheaper than catching it in production. I've also built and maintained the CI/CD pipelines that get all of it shipped.
Technical Leadership & Mentorship	I've been the technical lead on small projects, most recently a 2-3 engineer rewrite of a legacy system, and earlier a 5-6 person team where I also ran Scrum. I set direction through RFCs, PR checklists, and API design guides, and I sit on AMA's Architecture Steering Committee. The mentoring side matters to me too: helping newer developers find their way through legacy code and getting a team comfortable with TDD.
Cloud-Native Infrastructure	I work across production AWS spanning multiple accounts (Lambda, EKS, SQS, RDS, IAM, ECR), with Terraform handling the infrastructure as code. I deployed a production EKS cluster running cert-manager, Karpenter, and Linkerd, and moved a brittle legacy document system onto it as containerized microservices with Helm. Most of what I build now leans on Postgres and Datadog for the data and the visibility.

Technologies

Backend Languages	Java, Gosu (JVM), Python, TypeScript, C#, Node.js, Bash
Design & Practices	Object-Oriented Design, Design Patterns, TDD, Code Review, RFC Authorship
Architecture & APIs	RESTful APIs, OpenAPI, API Versioning, Microservices, Event-Driven Architecture
Databases	PostgreSQL, SQL, Aurora/RDS, Data Modelling, Redis
Cloud & Infrastructure	AWS (Lambda, EKS, EC2, RDS, S3, SQS, IAM, ECR), Terraform, Kubernetes, Docker, Helm
CI/CD & Automation	CI/CD, GitHub Actions, Jenkins, GitOps, Infrastructure as Code
Observability	Datadog, Elasticsearch, Kibana, Metrics Design, SLOs, Alerting